NLP

Embeddings Visual Guide

Turning words into vectors:
word2vec, analogies, cosine similarity, sentence embeddings.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Words as Vectors

Dense vectors carry meaning; one-hot vectors don't.

One-hot

50,000 dims, all zeros

but one 1. No notion of similarity.

Embedding

300 dense dims, learned from context. Similar words sit close.

Distributional hypothesis: words appearing in similar contexts get similar vectors. Geometry becomes semantics.

STEP 2/5

Skip-gram vs CBOW

Word2vec's two training games.

Skip-gram

predicts surrounding words. Better for rare words.

center word ->

CBOW

word. Faster, better for frequent words.

context -> center

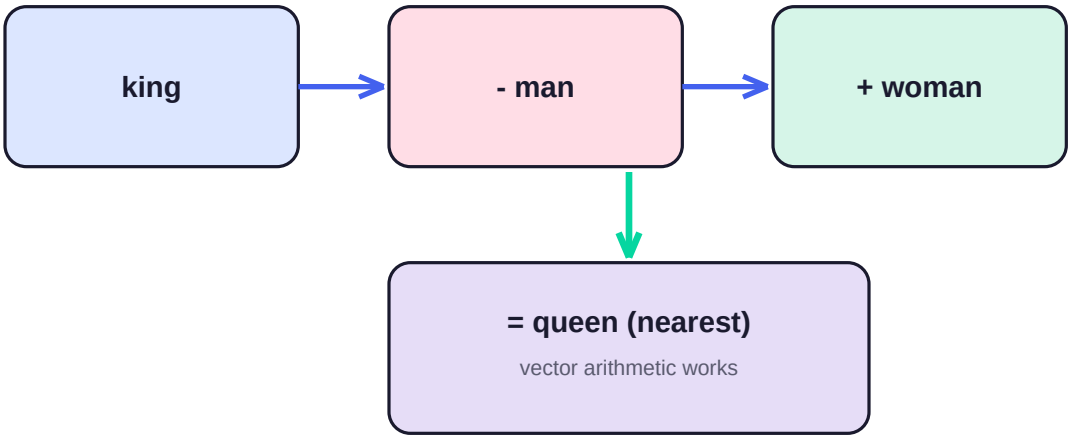
Both train a shallow network whose hidden weights BECOME the embeddings. The task is a pretext.

Negative sampling makes training tractable: contrast the true context word against a few random impostors.

STEP 3/5

Analogies in Vector Space

Relationships become directions.



king - man + woman lands nearest to 'queen'. Gender, tense, and capital-of become consistent offsets.

Analogies are a diagnostic, not a guarantee — they also surface the biases baked into the training corpus.

STEP 4/5

Cosine Similarity

Angle, not distance, measures relatedness.

cos = A dot B / (|A||B|)

1 = same direction

0 = unrelated

orthogonal vectors

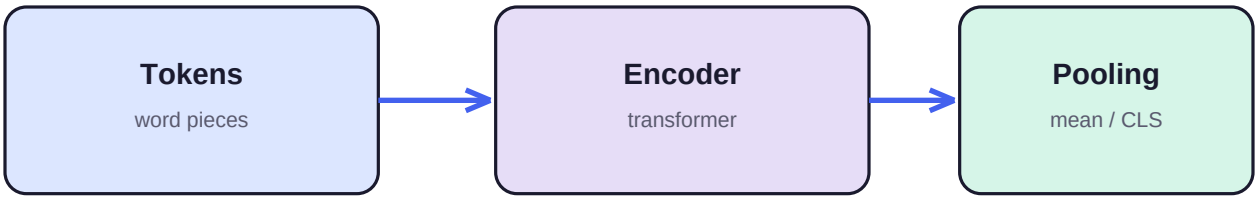
Cosine similarity ranges from -1 to 1. Normalizing vectors makes dot product equal cosine.

This one formula powers semantic search: embed the query, find stored vectors with the highest cosine similarity.

STEP 5/5

Sentence Embeddings

From word vectors to whole-text vectors.



- Mean-pool token embeddings, or train a dedicated sentence model (SBERT, E5, BGE).
- Contrastive training: pull paraphrases together, push unrelated texts apart.
- One vector per document unlocks semantic search, clustering, dedup, and recommendations.

STEP ✓/✓

Key Takeaways

Embeddings — Remember

- Dense vectors encode meaning; one-hot vectors cannot.
- Skip-gram predicts context from a word; CBOW predicts the word from context.
- Analogies (king - man + woman = queen) reveal semantic directions — and corpus bias.
- Cosine similarity turns angle into a relatedness score for search and clustering.
- Sentence embeddings power semantic search, dedup, and recommendations.

Keep exploring

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